

EXERCISES ON QUADRATIC RECIPROCITY

Remark: p will denote an odd prime unless otherwise stated.

Exercise 1. Compute the Legendre symbol $\left(\frac{51}{107}\right)$.

Exercise 2. Is there an odd prime p such that 7 is a quadratic residue $(\bmod p)$? If so, can you list 3 such primes?

Exercise 3. What are all the primes p such that $x^2 \equiv -3 \pmod{p}$ has a solution?

Exercise 4. Given that 2 is a primitive root $(\bmod 13)$, list all the quadratic residues $(\bmod 13)$.

Exercise 5. If n has a primitive root, how many incongruent $(\bmod n)$ primitive roots does it have? Express your answers in terms of the function ϕ .

Exercise 6. Recall Euclid's algorithm for the fact that there are infinitely many primes. Modify the same to show that there are infinitely many primes of the form $4k + 1$.

Exercise 7. Show that $\sum_{a=0}^{p-1} \left(\frac{a}{p}\right) = 0$ where left hand side is the sum over Legendre symbols.

Exercise 8. Does $x^2 \equiv 58 \pmod{77}$ have a solution?

Exercise 9. Show that $\sum_{a=1}^{p-2} \left(\frac{a(a+1)}{p}\right) = -1$, where the left hand side is the sum over Legendre symbols. Conclude that there is at least two consecutive integers $0 \leq i, i+1 \leq p$ such that both of them are quadratic residues.

Exercise 10. Show that the quadratic equation $ax^2 + bx + c \equiv 0 \pmod{p}$ has exactly $1 + \left(\frac{b^2 - 4ac}{p}\right)$ incongruent solutions $(\bmod p)$. (Here $\left(\frac{b^2 - 4ac}{p}\right)$ denotes the Legendre symbol, and not a ratio).

Exercise 11. For what primes q is -2 a quadratic residue $(\bmod q)$? Conclude that there are infinitely many primes of the form $8k + 3$.

Exercise 12. Consider the following generalization of the Legendre symbol due to Jacobi:

$$\left(\frac{a}{n}\right) := \left(\frac{a}{p_1}\right)^{\alpha_1} \left(\frac{a}{p_2}\right)^{\alpha_2} \cdots \left(\frac{a}{p_n}\right)^{\alpha_n}.$$

Show that it is possible such that $\left(\frac{a}{n}\right) = 1$ and yet $x^2 \equiv a \pmod{n}$ may not have a solution.

Exercise 13. Consider the previous exercise. Show that if $\left(\frac{a}{n}\right) = -1$, then $x^2 \equiv a \pmod{n}$ may not have a solution.

Exercise 14. Prove that there are infinitely many prime of the form $8k + 3$.

Exercise 15. For what p can we find (x, y) such that $x^2 + y^2 \equiv 0 \pmod{p}$? Can you find all the congruence classes for (x, y) , but we did not see last $(\bmod p)$, all such possible pairs (x, y) ?